

Psychobiotics & The Microbiomes That Make Us

Ryan Burns



Agenda

Introduction & Disclaimers

Science of the Gut Microbiome

Individual Case Study

What's Next

Q&A

Who I Am

- Data Scientist
- Type 1 Diabetic
- Disability Advisory Board Member
- Partner/Caregiver



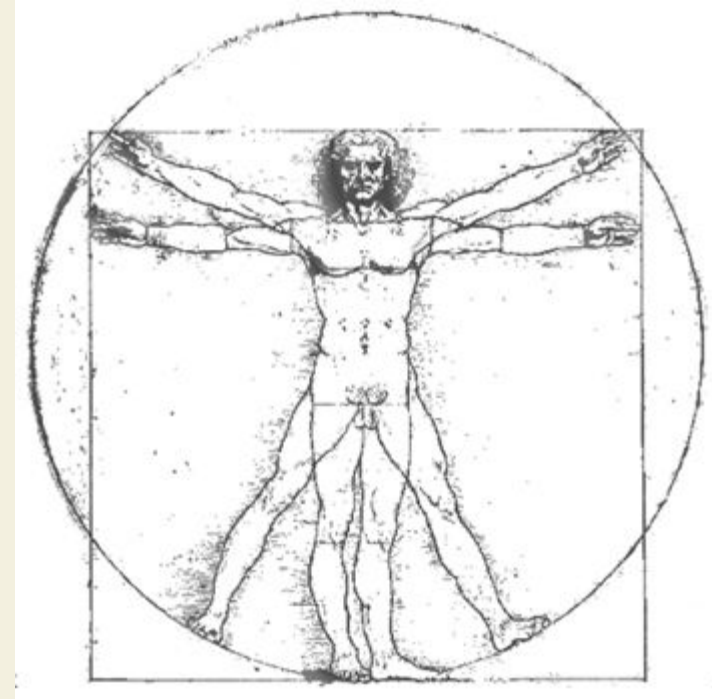
WHO I AM NOT

- Doctor
- Medical Professional

~~MD~~

Microbiomes & Gut Science

They're everywhere!!!



Human Microbiomes

Our Cells Are Outnumbered!

We contain roughly **57% microbial cells** (39 trillion prokaryotic cells) and **43% human cells** (30 trillion eukaryotic cells)¹

Microbial cells are much smaller than human cells

If it's exposed, it has a microbiome

- Skin
- Oral
- Lungs
- Gut
- Anal
- Vaginal

The Gut Microbiome

The total mass of your gut microbiota weighs roughly **2–3 pounds**, roughly **equal to the weight of the human brain**²

The microbes in our gut act as biochemical factories, fermenting dietary fiber to produce short-chain fatty acids (SCFAs) like acetate, propionate, and butyrate.

They also produce neurotransmitters! The gastrointestinal tract is responsible for producing:

- Approximately **90%–95% of the body's total serotonin**
- Roughly **50% of the body's total dopamine**

Vagus Nerve

Connection facilitating the “Gut-Brain Axis”

Vagal nerve endings don't cross into the gut, they receive chemical signals produced by our gut microbes

Sensory tool informing the brain about body functionality

- **80% of vagus nerve fibers are sensory**



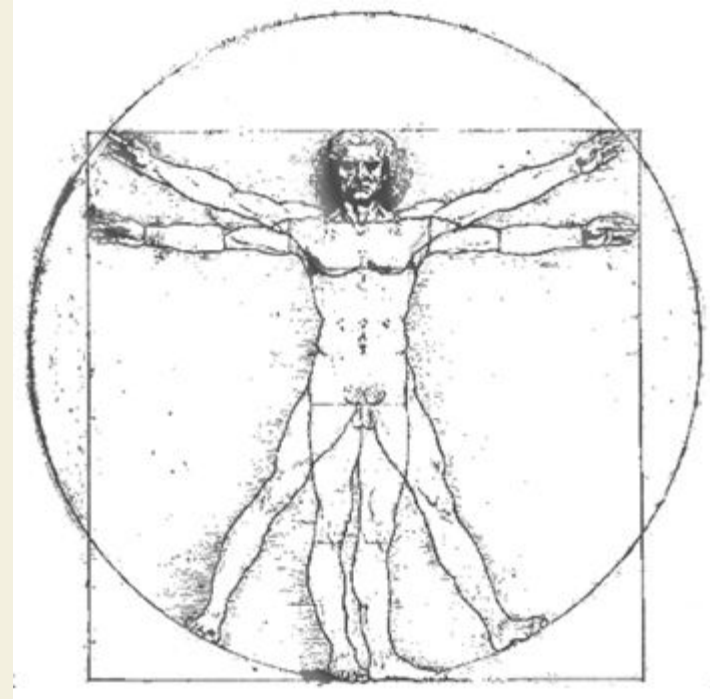
The Gut and Our Immune System

Gut contains 70–80% of our immune system

- GALT (Gut-Associated Lymphoid Tissue)
 - Primary training ground for immune cells
 - Where the outside world meets our internal bloodstream
- SCFAs promote the development of Regulatory T-cells
- Intestinal Permeability (Leaky Gut)
 - Can occur when beneficial microbes decrease, degrading the mucosal barrier of the gut
 - Allows bacterial components to leak into the circulatory system
 - Many bacteria have surface proteins similar to human cells, causing the immune system to create antibodies to attack human tissues (Molecular Mimicry)

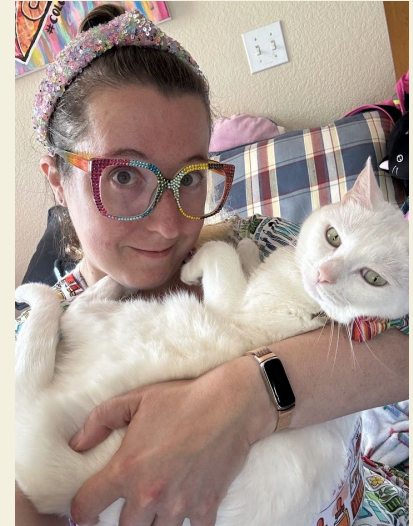
Case Study

Applying the science to treat bipolar disorder



Meet Monica!

Local artist, cat lover, Duke basketball fan, glitter scientist, color queen, rainbow enthusiast



Bipolar Disorder II

Illness burden is dominated by major depression with occasional bouts of hypomania

Distinct from Bipolar I but not a “milder” version in any sense, formally recognized in the DSM-IV in 1994

Bipolar disorder affects approximately 8 million adults in the US (2.4%) and approximately 40 million individuals worldwide (0.5%)

Life expectancy reduced by ~12–14 years

Higher prevalence of metabolic syndrome (37%), obesity (21%), type 2 diabetes (T2DM) (14%)

Approximately 15–20% of people with bipolar disorder die by suicide

- Annual rate 64x greater than general population (0.9% vs 0.014%)

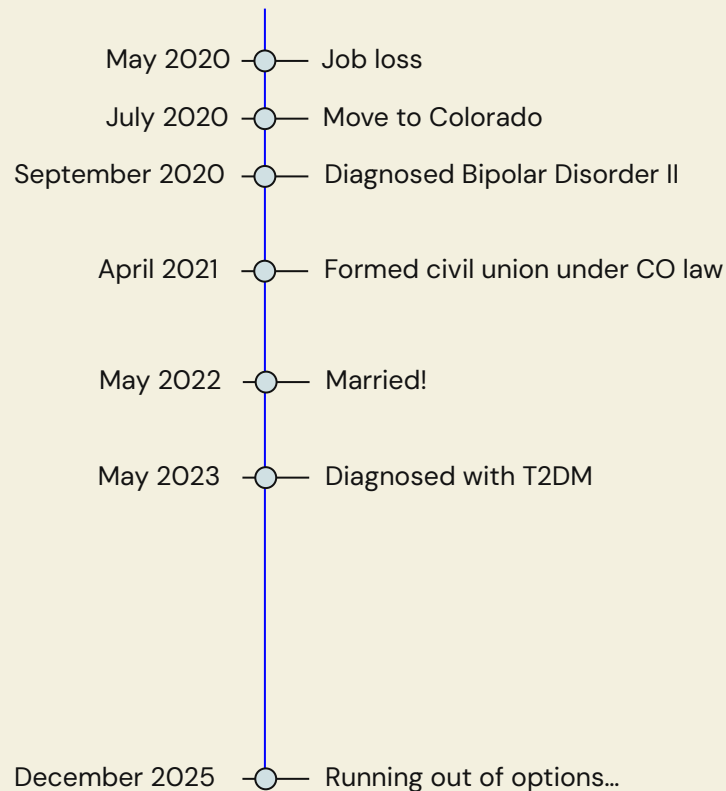
JAMA 2023³

Monica's Journey

Between 2020–2025, Monica tried most available pharmaceuticals for treating bipolar disorder (atypical antipsychotics and mood stabilizers)

Common link between metabolic and psychiatric illness: individuals with Bipolar Disorder have double the risk of T2DM

Some bipolar disorder medications can cause weight gain and insulin resistance



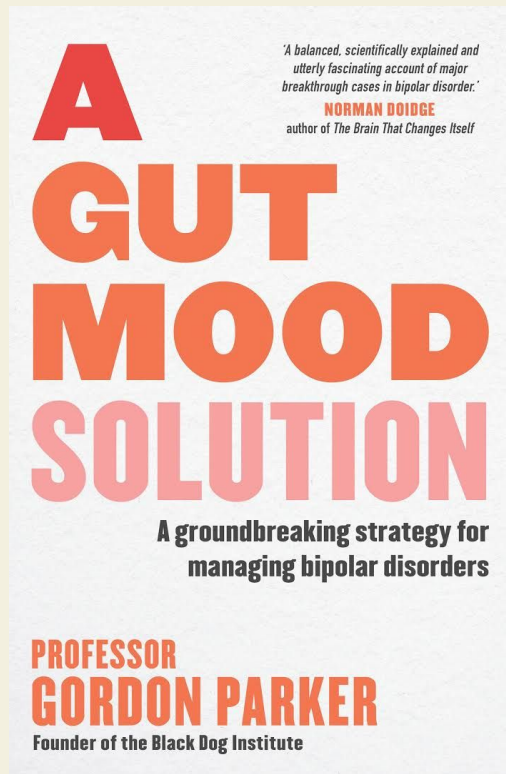
Meet Jane Dudley

Australian woman who treated her Bipolar Disorder I via Fecal Matter Transplant (FMT) in 2016

First read about her in 2023 or 2024

ABC News segment “Gut Instinct” aired July 2025⁴

Built her own personal website “Microbiome in Mind”⁵



Monica's Results with Fecal Matter Transplants

All Data Collected with eMoods App

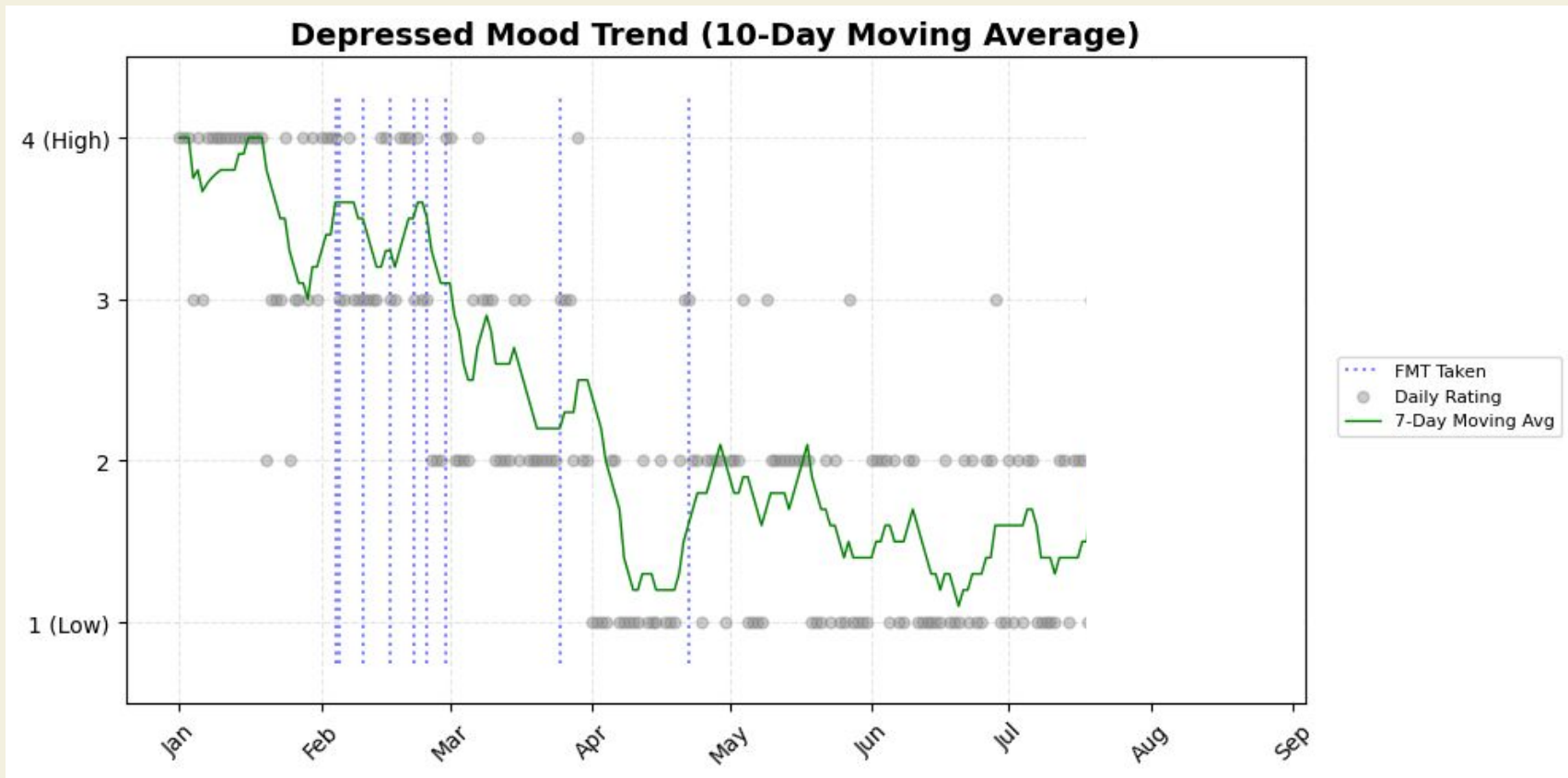
User-friendly app for patients to track symptom data relating to Bipolar I and II disorders, Depression, PTSD, and Anxiety Disorders

<https://emoodtracker.com/>

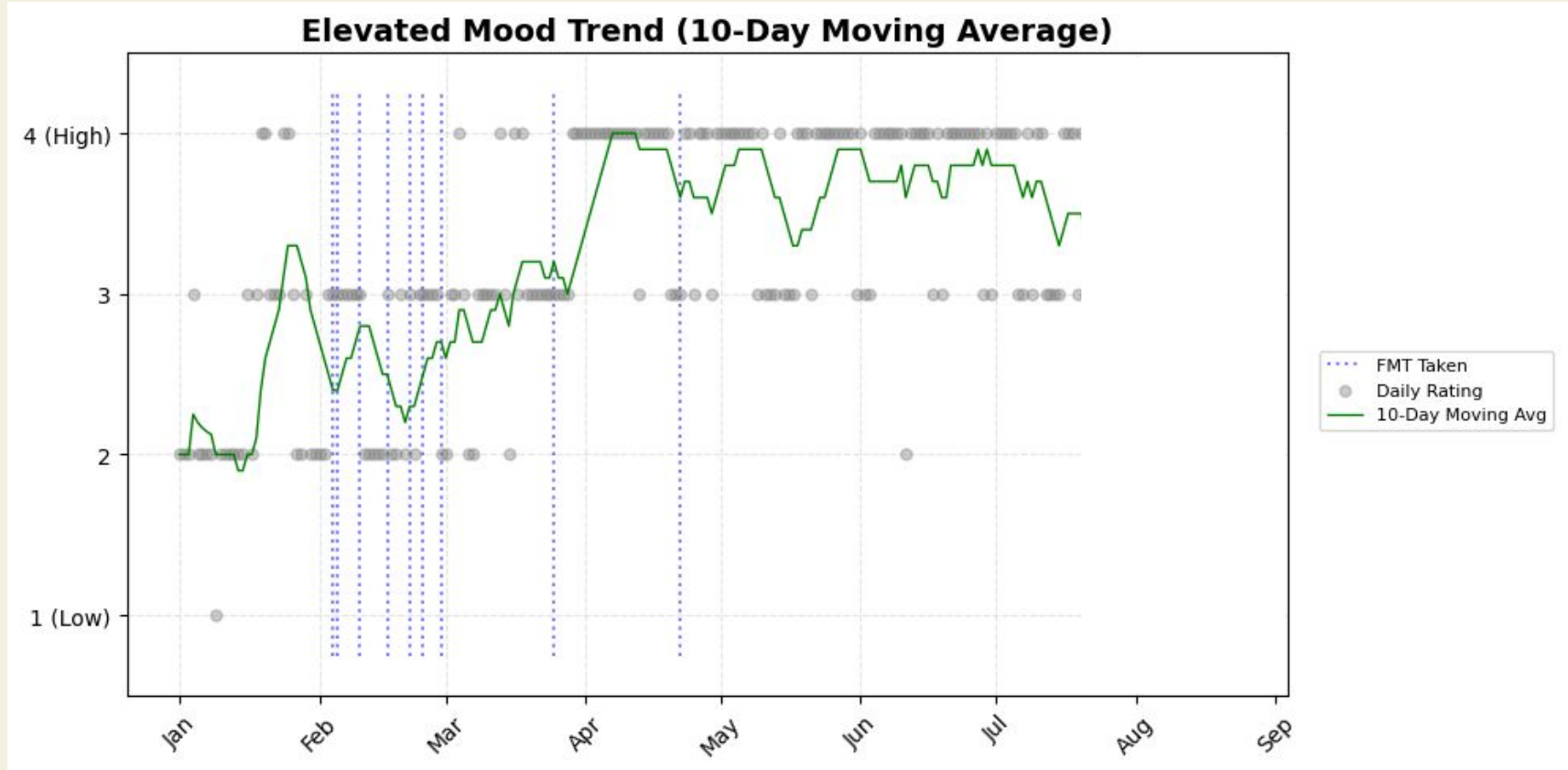
The screenshot displays the eMoods App interface. At the top, there is a date selector with 'Today' on the left, navigation arrows, 'Thu, Aug 20' in the center, and a share icon on the right. Below this is a section for 'Hours Slept Last Night: 0.0' with a horizontal slider and minus/plus buttons. The app then presents four scales, each with a title and four response options: 'None', 'Mild', 'Moderate', and 'Severe'. The scales are: 'Today's Most Extreme Depressed Mood', 'Today's Most Extreme Elevated Mood', 'Today's Most Extreme Irritability', and 'Today's Most Extreme Anxiety'.

Scale Title	None	Mild	Moderate	Severe
Today's Most Extreme Depressed Mood				
Today's Most Extreme Elevated Mood				
Today's Most Extreme Irritability				
Today's Most Extreme Anxiety				

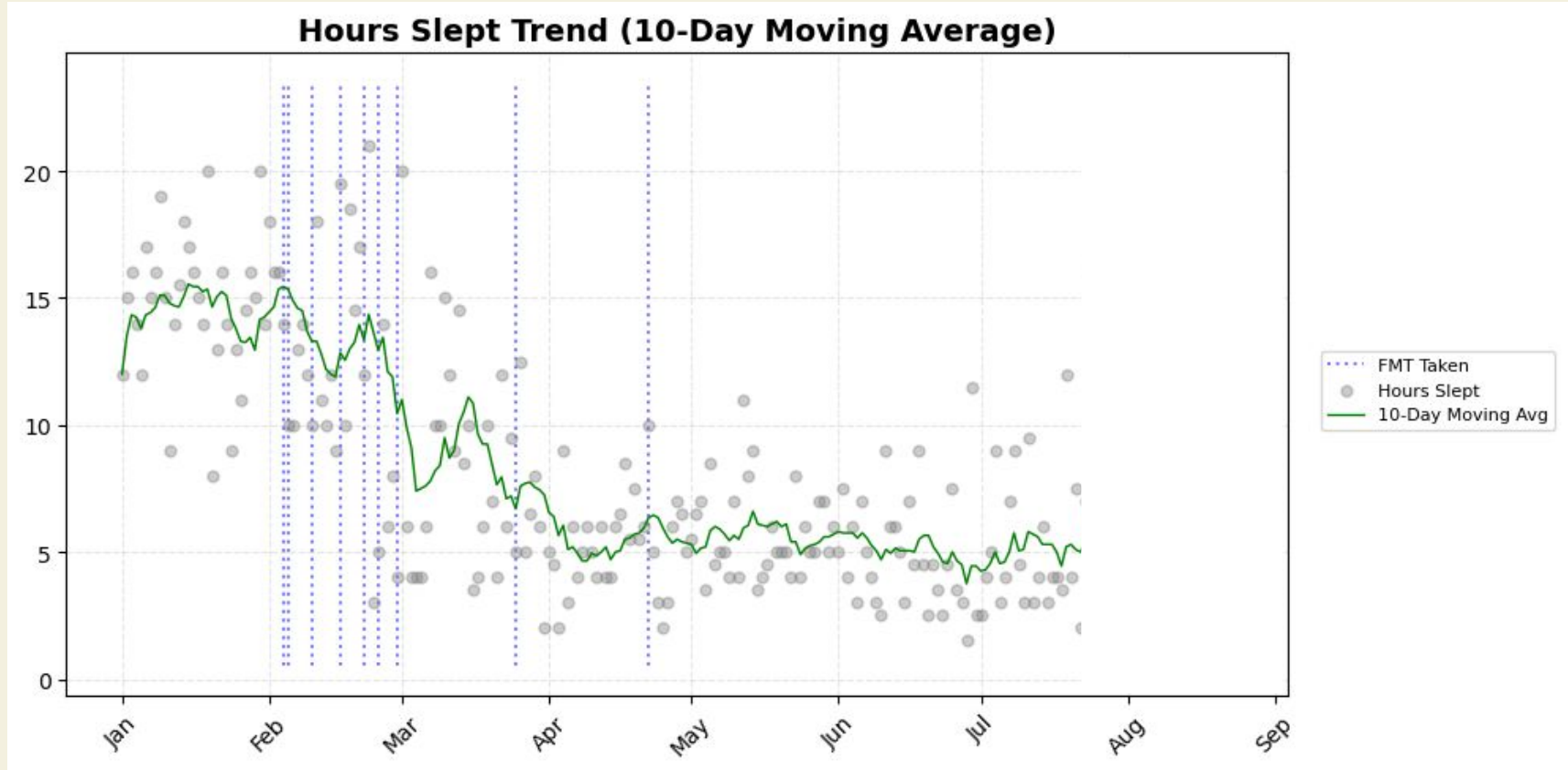
Monica's Results with FMT – Depressed Mood

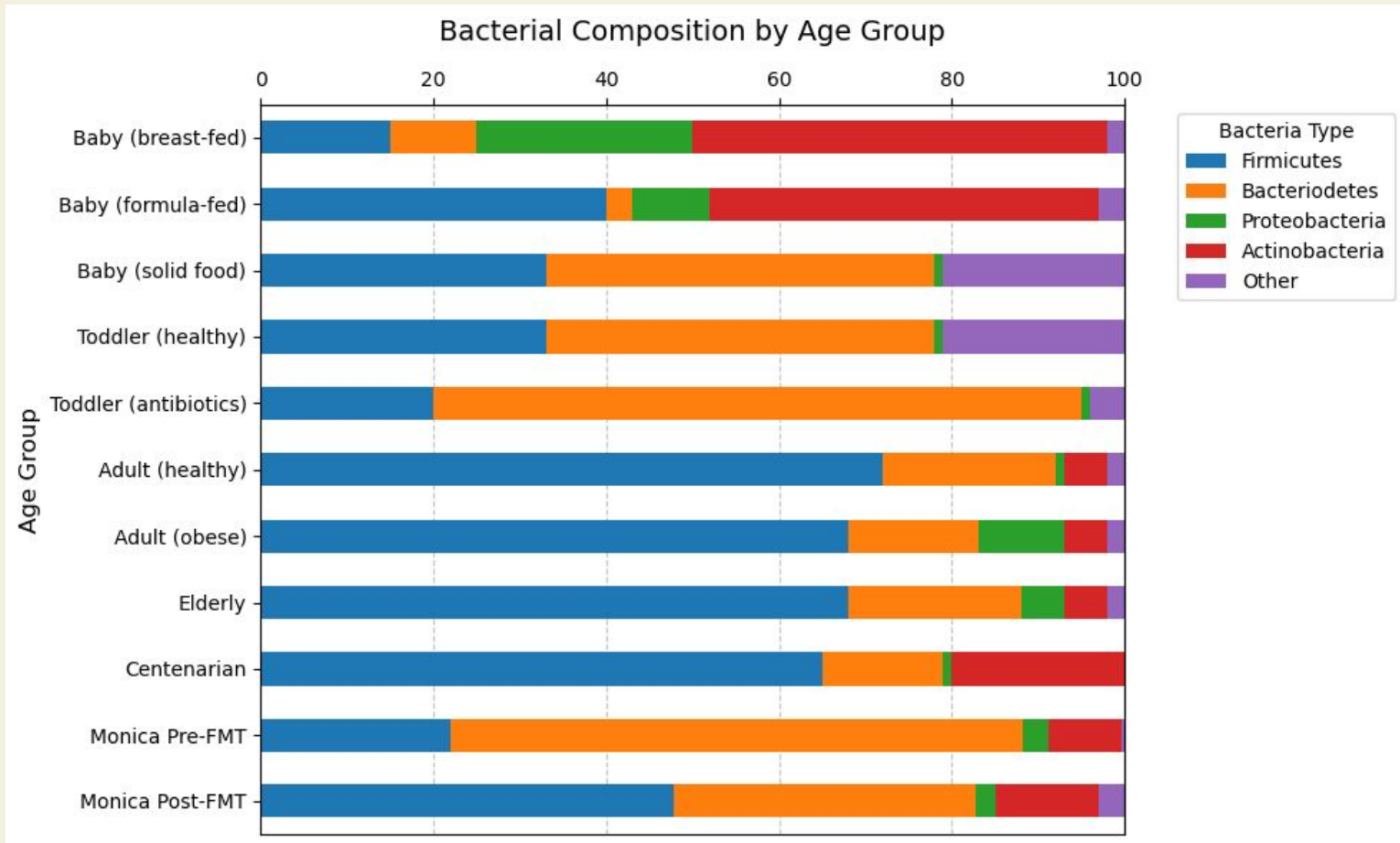


Monica's Results with FMT – Elevated Mood



Monica's Results with FMT – Hours Slept





What's Next?

Outstanding Inquiries & Challenges

Further Reading & Conversation

Future of Gut Microbiome Treatments

- Mental Health Treatments
 - Bipolar Disorder
 - Major Depressive Disorder
 - Schizophrenia
 - A study induced schizophrenic behavior in mice using FMT from human schizophrenic donors⁶
- Autoimmune Treatments
 - Multiple Sclerosis (MS)
 - 15 adult patients received FMT for gastrointestinal symptoms, all experienced improved and reversed neurological symptoms secondary to MS⁷
 - Ulcerative Colitis & Crohn's (Inflammatory Bowel Disease)
 - In a double-blind, multi-center RTC, 27% (11/41) of patients receiving intensive FMT achieved steroid-free clinical remission with endoscopic healing, compared to only 8% (3/40) in the placebo group⁸
 - Rheumatoid Arthritis (RA) & Psoriatic Arthritis
 - Alopecia Areata / Alopecia universalis
 - Two patients with alopecia universalis developed subsequent hair regrowth after FMT for treatment of recurrent *C. difficile* infections⁹
 - Type 1 Diabetes
 - Faecal microbiota transplantation halts progression of human new-onset type 1 diabetes 12 months after disease onset, preserving residual beta cell function¹⁰

Challenges

High variation in gut microbiome

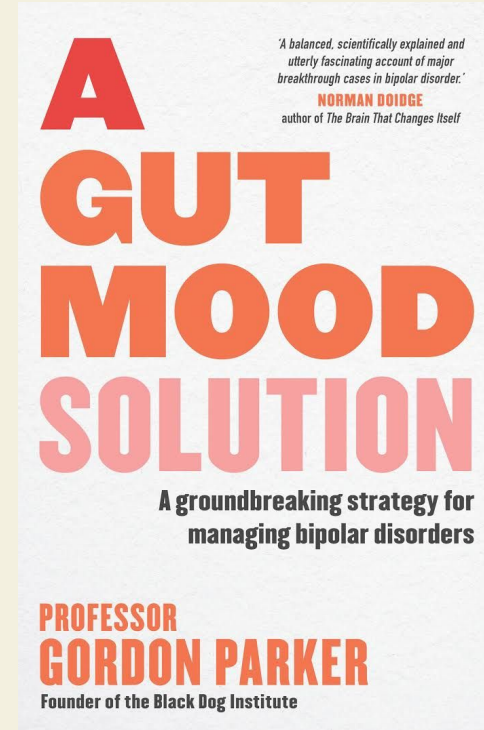
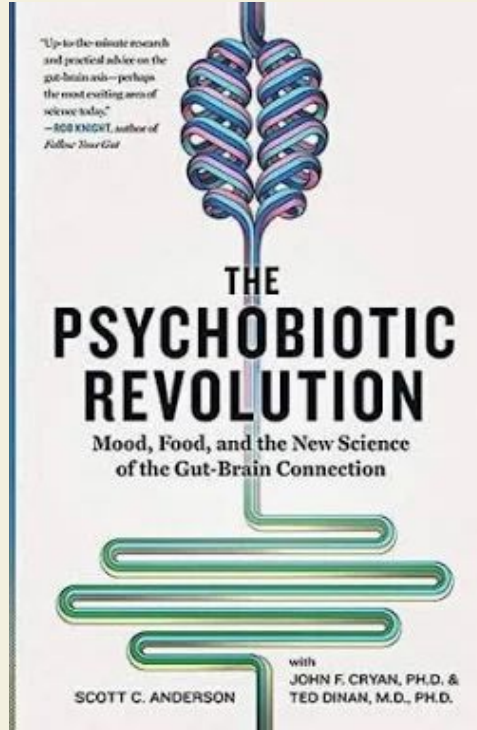
High dietary/environmental impacts

FMT Safety

Healthcare cost & coverage

Mental health stigma

Further Reading



Sources

- 1 - <https://www.sciencefocus.com/the-human-body/human-microbiome>
- 2 - <https://www.ncbi.nlm.nih.gov/books/NBK562894/>
- 3 - <https://jamanetwork.com/journals/jama/article-abstract/2810502>
- 4 - abc.net.au/news/2025-07-28/gut-instinct-jane-dudley/105583270
- 5 - <https://www.microbiomeinmind.com.au/>
- 6 - The gut microbiome from patients with schizophrenia modulates the glutamate-glutamine-GABA cycle and schizophrenia-relevant behaviors in mice <https://pubmed.ncbi.nlm.nih.gov/30775438/>
- 7 - The Potential Role of Fecal Microbiota Transplant in the Reversal or Stabilization of Multiple Sclerosis Symptoms: A Literature Review on Efficacy and Safety <https://pmc.ncbi.nlm.nih.gov/articles/PMC10745313/>
- 8 - Multidonor intensive faecal microbiota transplantation for active ulcerative colitis: a randomised placebo-controlled trial <https://pubmed.ncbi.nlm.nih.gov/28214091/>
- 9 - Hair Growth in Two Alopecia Patients after Fecal Microbiota Transplant <https://pmc.ncbi.nlm.nih.gov/articles/PMC5599691/>
- 10 - Faecal microbiota transplantation halts progression of human new-onset type 1 diabetes in a randomised controlled trial <https://pubmed.ncbi.nlm.nih.gov/33106354/>